MABEL: Verified Bill of Materials

A line-item cost model for an open mobile bimanual manipulator

MABEL project

31 July 2026

Abstract

This document is the audited bill of materials (BOM) for MABEL, an open-source mobile bimanual manipulator with a three-module swerve base, two 7-DOF quasi-direct-drive arms, two tendon-driven ORCA hands, an actuated lift–torso–neck body, and a five-camera plus LiDAR perception suite — 56 functional degrees of freedom driven by 59 motors. Every line of a purchase record was re-priced against a vendor page or catalog listing on 2026-07-31, cross-checked against what the repository's firmware and hardware-abstraction layer say is actually installed, and converted to a single currency at one stated exchange rate. The as-built platform, including an NVIDIA Jetson AGX Thor developer kit, costs **$12,874**. Swapping Thor for a Jetson Orin Nano Super brings the same robot to **$9,624** — $172 per functional degree of freedom, and below the $10,000 threshold the project claims — though that margin does not survive the uncosted items of §5. Without onboard compute the platform is $9,375; without compute or perception, $7,852. Verification lowered the total by only $233.11 against the submitted sheet, but that small net figure hides four substantive corrections, including a swerve-module line that omitted $847.50 of motors and controllers and a neck actuator listed as the wrong part. We also report what the sheet does *not* cover ($510–$1,255 of uncosted structure, PCBs and hardware), because a BOM that hides its gaps is not a BOM.

Keywords: bill of materials; open-source robot hardware; mobile manipulation; cost analysis; reproducible robotics; design for replication.

1 Overview and scope

MABEL's central hardware claim is that a research-grade mobile bimanual robot with anthropomorphic hands and an articulated neck can be built for under $10,000 from catalog parts, with custom fabrication limited to bent laser-cut sheet metal, printed covers, and two-layer distribution PCBs. That claim is only worth as much as the BOM behind it. This document is that BOM.

What is in scope. Every physical part required to build one MABEL: actuators, structure, fasteners, sensing, onboard networking, power, embedded controllers, and onboard compute. Prices are catalog list or as-purchased unit prices, excluding shipping, duties, sales tax, tools, and consumables.

What is out of scope. Labour, the teleoperation headset (an Apple Vision Pro or an iPhone is a client, not a robot part), the development workstation, and training compute. Failure spares are discussed in §7 but not costed.

Source of truth. The line items live in `BOM/data/mabel_bom.csv`. Every number, table and figure in this document — and the JSON the project website reads — is regenerated from those CSVs by `BOM/tools/build_bom.py` (§6). Nothing here is typed by hand.

2 Method

2.1 Currency

Chinese-vendor parts (the DAMIAO actuators, the OpenArm structure, the Feetech hand servos and their bus boards) are quoted in CNY. All are converted at a single rate,

$$1 \text{ USD} = 6.75 \text{ CNY} \qquad \text{(spot, 2026-07-31)}, \tag{1}$$

taken from the Federal Reserve H.10 release and cross-checked against commercial spot quotes for the same date (6.7498), then rounded to two figures. The rounding error over the entire CNY-denominated block ($4,328) is under $5. Using a single dated rate rather than each part's historical purchase rate makes the document reproducible: re-running the build script with a new rate moves every CNY line coherently. The submitted purchase sheet implied a rate near 6.97; at that rate the CNY block would be $138 cheaper, which is the sensitivity of this BOM to the exchange rate.

2.2 Verification protocol

Each of the 57 lines was assigned one of four states.

verified The unit price was read off the vendor's own product page on 2026-07-31 (REV Robotics, ROBOTIS, SparkFun, NVIDIA).

sheet Taken from the purchase record without independent re-pricing. This applies to Taobao listings, which are not stably addressable, and to Amazon/marketplace listings whose prices float. These are honest as-purchased prices, not catalog prices.

corrected The sheet and the evidence disagreed, and the line was changed. Marked † in Table 3 and itemised in §4.

added The part is demonstrably on the robot — the firmware or the hardware-abstraction layer names it — but was missing from the sheet. Marked ‡.

Two independent evidence channels were used. *Vendor evidence* settles price and what a kit contains. *Repository evidence* settles what is actually installed: the hardware-abstraction layer identifies servos by the model number they report over the bus, the firmware manifest names each microcontroller and its bus, and the MuJoCo model is the canonical geometry. When a marketing page and the code disagreed, the code won — code that talks to a device is a stronger witness than a page describing an intent.

3 Results

3.1 Cost by subsystem

3.2 Build scenarios

The compute choice dominates the headline number, so it is reported as a scenario rather than folded into a single figure.

S1 — as built ($12,874). Everything in Table 3, including the Jetson AGX Thor developer kit at $3,499. Thor is the right choice for onboard VLA-scale inference and is what the physical robot runs today, but it is 27% of the build cost and is the least MABEL-specific part of it.

Subsystem	Cost ($)	Share
Mobile base	1,920	14.9%
Body	899	7.0%
Arms	2,709	21.0%
Hands	1,234	9.6%
Head & neck	296	2.3%
Sensing	1,523	11.8%
Structure & fasteners	184	1.4%
Electronics & power	611	4.7%
Compute	3,499	27.2%
Total	**12,874**	**100.0%**

Table 1: As-built cost by subsystem. Onboard compute is the single largest line at 27% of the total; actuation and structure together (base, body, arms, hands, neck) are $7,056 or 55%.

Scenario	What it includes	Total ($)
S1 As built	every line below, including the Jetson AGX Thor dev kit	12,874
S2 Budget compute	Thor swapped for a Jetson Orin Nano Super (8 GB)	9,624
S3 Platform, no onboard compute	driven from a tethered laptop or an existing host	9,375
S4 Actuation + structure only	no compute, no cameras, no LiDAR	7,852

Table 2: Total cost under four build scenarios. S2 (highlighted) is the recommended replication configuration and the number the project's "sub-$10k" claim should be attached to.

S2 — budget compute ($9,624). The same robot with a Jetson Orin Nano Super (8 GB, $249) in place of Thor: a $3,250 reduction, or 25% of the total, for one substitution. The Orin Nano Super delivers 67 TOPS against Thor's Blackwell-class throughput, so it comfortably runs the ROS 2 stack, SLAM, navigation, the 50 Hz state stream and the video pipeline, and it trains nothing. It is the right configuration for a teleoperation and data-collection platform where policies are trained off-board and, if run onboard at all, are run small. This is the honest "build one yourself" number.

S3 — platform, no onboard compute ($9,375). The robot driven from a tethered laptop or a host the lab already owns. Everything except the Jetson: all actuation, all structure, all sensing, all networking and power.

S4 — actuation and structure only ($7,852). No compute, no cameras, no LiDAR: a 56-DOF teleoperable mobile bimanual body. This is the mechatronics floor and the number to compare against arm-only or base-only kits.

3.3 Cost in context

Why the cost is low — five specific reasons, not one.

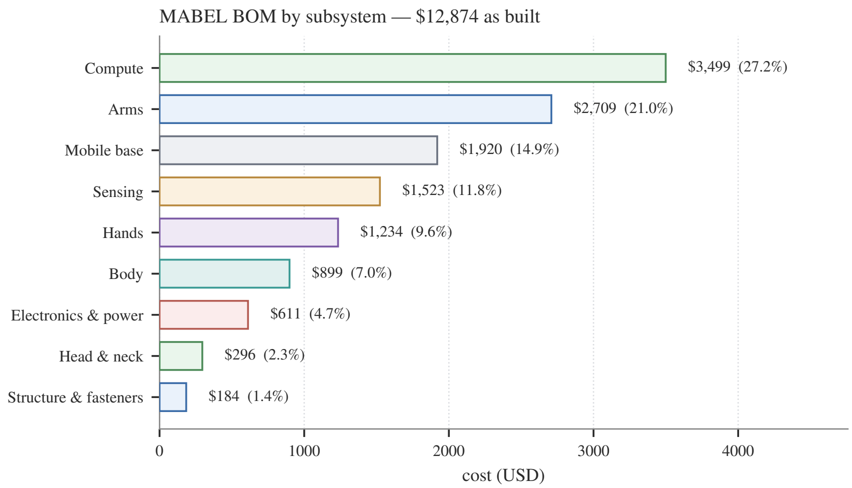

Figure 1: Subsystem cost breakdown of the as-built robot. The arms are the most expensive mechanical subsystem ($2,709, 21%), driven almost entirely by the fourteen quasi-direct-drive actuators; the swerve base follows once its motors and controllers are counted honestly.

1. **The expensive parts are commodity, and the specialised parts are cheap.** Quasi-direct-drive actuators reached commodity pricing through the quadruped market: fourteen arm actuators cost $2,181, where a comparable set of harmonic-drive joints with dual encoders is typically several times the price. The parts that are genuinely MABEL-specific — brackets, plates, covers — are the *cheapest* lines in the BOM ($635 for every plate, panel, extrusion and fastener on the robot), because they are flat patterns and printed parts rather than machined assemblies.

2. **Three swerve modules instead of four.** A three-module holonomic base gives the same three planar degrees of freedom as a four-module base and removes one module, one drive motor, one steering motor and two motor controllers: $557.50 saved, at the cost of a smaller support polygon that the tip-over model has to respect.

3. **Dexterity comes from servo count, not servo cost.** Thirty-four Feetech bus servos give 34 actuated hand DOF for $1,234. A pair of commercial dexterous hands at comparable DOF is a five-figure line on its own; the tendon-driven printed-frame approach converts that into $36 per actuated hand joint plus maintenance labour.

4. **No institutional machine shop.** Every custom part is a bent laser-cut flat pattern, a printed cover, or a two-layer PCB. That is a pricing decision as much as a fabrication one: it caps the custom fabrication content of the robot at a few hundred dollars and makes the design replicable by anyone with a printer and a sheet-metal vendor.

5. **Compute is a choice, not a fixture.** Because everything that moves the robot is gated behind one control library and one wire contract, and because the heavy learning workloads run off-board, the onboard computer is swappable. That is what makes the $3,250 difference between S1 and S2 available at all.

At $172 per functional degree of freedom (S2), MABEL is priced comparably to single-arm research kits while providing a mobile base, a second arm, two dexterous hands, and an actuated neck.

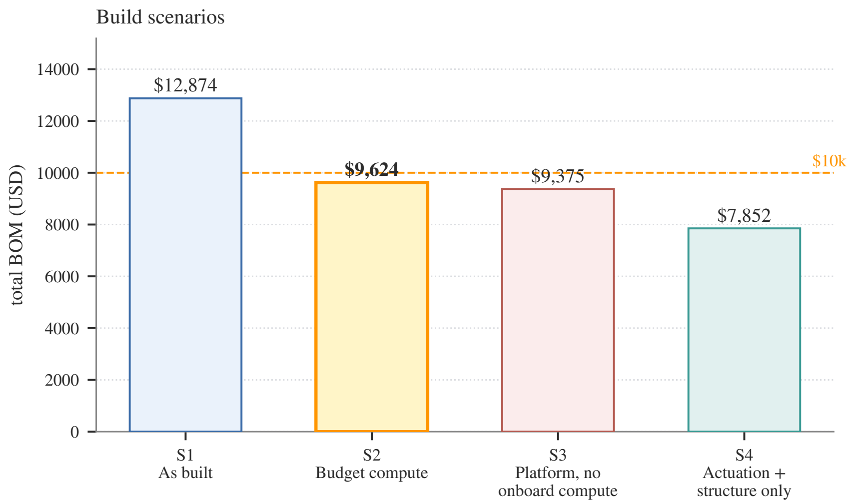

Figure 2: Scenario totals against the $10,000 line. Only the Thor developer kit pushes the platform over it; the mechanics, sensing and electronics of a 56-DOF mobile bimanual robot fit under $9.4k on their own.

3.4 The itemised BOM

Table 3 is the complete list. Unit prices marked ¥ were quoted in CNY and converted by Eq. 1; the printed value is already in USD. † marks a line corrected against the submitted sheet, ‡ a line added because the repository proves the part is installed.

Table 3: Complete itemised bill of materials. † corrected against the submitted purchase sheet; ‡ added because the repository proves the part is installed; ¥ quoted in CNY and converted by Eq. 1.

Item	Part no.	Qty	Unit ($)	Ext. ($)
Mobile base				
3 in MAXSwerve module†	REV-21-3005	3	275.00	825.00
NEO brushless motor V1.1 (drive)‡	REV-21-1650	3	42.50	127.50
NEO 550 brushless motor (azimuth)‡	REV-21-1651	3	30.00	90.00
SPARK Flex motor controller (drive)‡	REV-11-2159	3	110.00	330.00
SPARK MAX motor controller (azimuth)‡	REV-11-2158	3	100.00	300.00
Base sheet-metal panel 1	custom	1	51.75	51.75
Base sheet-metal panel 2	custom	1	195.36	195.36
subtotal				**1,919.61**
Body				
FlexiSpot E7 Pro lift column	E7 Pro	1	399.00	399.00
Lift-to-torso adapter plate	custom	1	59.26	59.26
Torso output bracket	custom	1	55.16	55.16
Body bottom plate	custom	1	33.55	33.55
Arm mounting plate	custom	2	13.17	26.34
Neck mounting plate	custom	1	29.05	29.05
DAMIAO DM-J10422P torso-pitch actuator	DM-J10422P	1	296.15 ¥	296.15
subtotal				**898.51**
Arms				
DAMIAO DM-J4310 actuator (wrist joints)†	DM-J4310	6	88.74 ¥	532.44

Table 3 – continued

Item	Part no.	Qty	Unit ($)	Ext. ($)
DAMIAO DM-J4340 actuator (shoulder roll + elbow)†	DM-J4340	4	130.72 ¥	522.86
DAMIAO DM-J8009P actuator (shoulder pitch/yaw)†	DM-J8009P	4	281.33 ¥	1,125.33
OpenArm structural parts (7075 CNC + sheet metal)	custom	1	399.85 ¥	399.85
Arm cable and connector harness†	custom	1	128.15 ¥	128.15
subtotal				**2,708.64**
Hands				
Feetech HLS3915M bus servo (finger)	HLS3915M	32	34.81 ¥	1,114.07
Feetech HLS3930M bus servo (wrist)	HLS3930M	2	53.19 ¥	106.37
Feetech TTL bus driver board	ST/SC series	2	4.10 ¥	8.19
Feetech 5264 power-distribution board	ST/SC series	2	2.67 ¥	5.33
subtotal				**1,233.97**
Head & neck				
DAMIAO DM-J4310 actuator (neck yaw)	DM-J4310	1	88.74 ¥	88.74
ROBOTIS Dynamixel XC330-T181-T (neck pitch + roll)†	XC330-T181-T	2	103.39	206.78
subtotal				**295.52**
Sensing				
Intel RealSense D405 depth camera (wrist)	82635DSD405	2	288.00	576.00
Intel RealSense D435i depth camera (front/chest)	82635D435IDK5P	1	319.72	319.72
Stereolabs ZED Mini stereo camera (head)	ZED Mini	1	399.00	399.00
Slamtec RPLIDAR A2M12 2-D scanner	A2M12	1	228.63	228.63
subtotal				**1,523.35**
Structure & fasteners				
Aluminum extrusion cut set	2x2 in and 2x4 in	1	101.46	101.46
Metric screw assortment	-	1	13.99	13.99
Imperial screw assortment (lift + extrusion)	-	1	8.99	8.99
Extrusion 90-degree corner brackets	-	1	26.90	26.90
Extrusion angled gusset plates (1010 series)	-	1	20.88	20.88
Magnets (cover retention)	-	1	11.99	11.99
subtotal				**184.21**
Electronics & power				
Buck converter 24 V to 12 V	-	1	18.99	18.99
6-port 2.5 GbE managed switch (4x2.5G + 2x SFP+)	SODOLA	1	49.99	49.99
SFP+ twinax DAC cable	-	1	45.99	45.99
USB expansion card / powered hub	-	1	74.99	74.99
Raspberry Pi Pico (lift controller)	-	1	15.95	15.95
BTS7960 H-bridge (lift motor)	-	1	9.99	9.99
Teensy 4.1 (swerve base controller)‡	Teensy 4.1	1	31.50	31.50
Teensy 4.1 (arm/torso/neck CAN controller)‡	Teensy 4.1	1	31.50	31.50
Ethernet patch cable	-	2	6.99	13.98
USB-C cable 10 ft (ZED Mini)	-	1	20.68	20.68
USB-C to USB-A cable	-	4	6.99	27.96
Micro-B to USB-A cable (wrist cameras)	-	2	13.99	27.98
Fuse box / blade fuse block	-	1	15.99	15.99

Table 3 – continued

Item	Part no.	Qty	Unit ($)	Ext. ($)
Cable sleeving 1/4 in	-	1	8.99	8.99
Cable sleeving 1/2 in	-	2	8.99	17.98
Battery pack	-	2	46.81	93.62
Main power cable (6 AWG XT-series)	-	1	24.99	24.99
USB sound card + 8 ohm 5 W speaker	-	1	19.50	19.50
Work light	-	1	7.99	7.99
Power cord extension	-	1	16.95	16.95
Wi-Fi antenna extension cable	-	2	17.99	35.98
subtotal				**611.49**
Compute				
NVIDIA Jetson AGX Thor Developer Kit	-	1	3,499.00	3,499.00
subtotal				**3,499.00**
Total, as built				**12,874.30**

4 Verification findings

4.1 Priced corrections

Four lines changed in value. The net effect is small — $233.11 against a $13,107 sheet — but the cancellation is coincidental, and each correction matters on its own.

Line	Sheet ($)	Verified ($)	Δ	Why
Swerve modules (3x)	2,025.00	1,672.50	−352.50	Sheet carried one $675 lump per module. REV-21-3005 is $275 and excludes motors and controllers, so the module is re-costed as module + NEO + NEO 550 + SPARK Flex + SPARK MAX = $557.50 each.
Neck Dynamixel servos	149.39	206.78	+57.39	Sheet listed 1x 2XL430-W250-T. The HAL reads model 1090 = XC330-T181 and config.py:665 says two of them, so 2x XC330-T181-T at $103.39.
Base and arm MCUs	0.00	63.00	+63.00	2x Teensy 4.1 (swerve base, arm/torso/neck CAN) are documented in firmware/CLAUDE.md but were not on the sheet.
Onboard compute	3,500.00	3,499.00	−1.00	Jetson AGX Thor Developer Kit MSRP is $3499.
Net	**5,674.39**	**5,441.28**	**−233.11**	

Table 4: Priced corrections, submitted sheet to verified BOM.

The swerve line was a lump, not a part. The sheet carried $675 per module against the REV MAXSwerve product page. That page prices the module at $275 and states that motors, motor controllers and framing are *not* included. A MAXSwerve module does not drive anything on its own: it needs a NEO drive motor, a NEO 550 azimuth motor, and a controller for each. Once those are itemised, the module costs $557.50 and the base line is $1,672.50 rather than $2,025. The

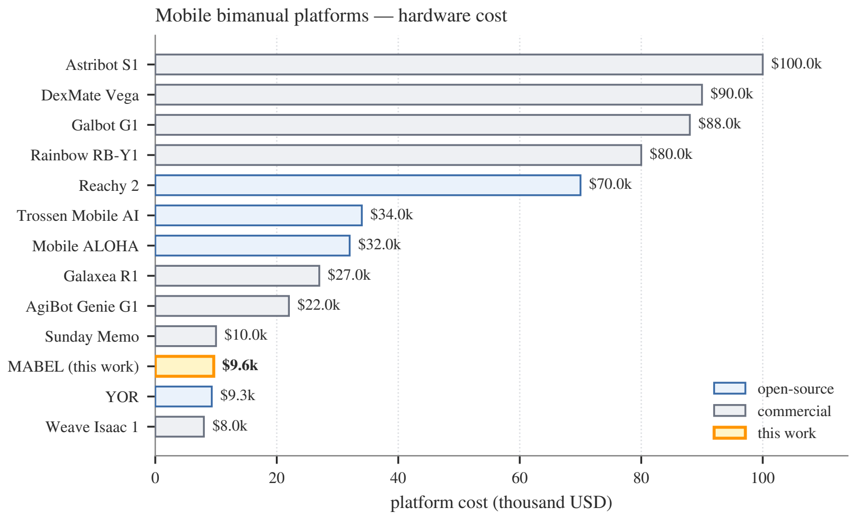

Figure 3: MABEL (S2) against surveyed mobile bimanual platforms; peer prices are manufacturer-stated or retail-listed as recorded in the project's platform survey. MABEL is cheaper than every commercial platform in the survey by a factor of 2–10, and is the least expensive open platform that has both multi-DOF hands and an articulated neck.

correction is worth making even though it reduces the total, because a single-line $675 lump is not reproducible — a reader ordering from it receives three modules that cannot move.

The neck servo was the wrong part. The sheet listed one ROBOTIS 2XL430-W250-T (a dual-axis servo) at $149.39. The hardware-abstraction layer disagrees, and it disagrees with evidence: the Dynamixel driver maps the model number the servos report over the bus, and its comment records that MABEL's neck reports model 1090, an XC330-T181; the hardware configuration states that there are two of them, one for pitch and one for roll. The BOM therefore carries 2× XC330-T181-T at $103.39. If a 2XL430 is in fact fitted to the physical robot, the driver's identification table is stale and should be corrected in code — either way, the two sources cannot both be right, and this is flagged as the one open question in this document.

Two microcontrollers were missing. The sheet costs a Raspberry Pi Pico for the lift but no Teensy. The firmware manifest names two Teensy 4.1 boards: one runs the swerve base, taking a body twist over UDP and driving the REV SPARK CAN bus; the other runs the arms, torso and neck at 500 Hz across three DAMIAO MIT-mode CAN buses. Both are load-bearing — without them the robot has no low-level control at all — and both are now costed.

Compute was $1 optimistic. The Jetson AGX Thor developer kit MSRP is $3,499.

4.2 Naming corrections (no price change)

Three actuator part numbers on the sheet do not exist as written; in each case the linked vendor listing and the repository's own build table identify the real part. `DMJ4301` → **DM-J4310** (six, arm wrists); `DMJ4343` → **DM-J4340** (four, shoulder roll and elbow); `DMJ4341` → **DM-J8009P** (four, shoulder pitch and yaw, 40 N m). The counts are right and total 14 actuators for two 7-DOF arms;

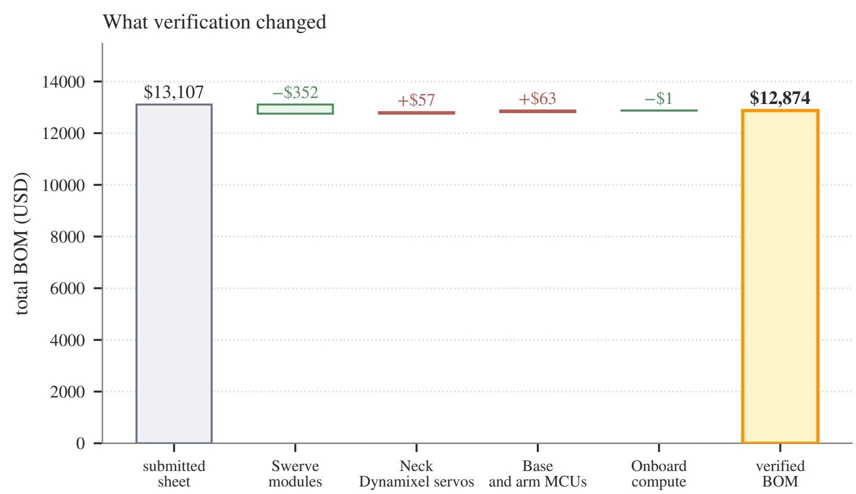

Figure 4: Waterfall from the submitted sheet to the verified BOM. The swerve-module correction dominates and is partly offset by the neck servos and the two microcontrollers the sheet had omitted.

only the labels were wrong. Ordering from the sheet as written would have produced three wrong parts, which is exactly the failure mode a verified BOM exists to prevent.

One further ambiguity is resolved by assumption rather than by evidence: the arm cable and harness line reads "865" with no currency against a Taobao vendor. It is read as ¥865 ($128). If it was in fact $865, the arms subtotal and every total in this document rise by $737.

4.3 Documentation drift found along the way

Verifying parts against the repository surfaced four published claims that no longer match the robot. They are recorded here because a BOM audit is the natural moment to catch them.

1. The public build guide lists an **RPLIDAR A3**; the installed scanner is an **A2M12**.
2. The build guide lists **Arducam global-shutter wrist cameras**; both wrists have carried **RealSense D405** units since July 2026, which the camera configuration confirms.
3. The build guide lists the neck's non-yaw axes as two **XC330-T181** servos, which agrees with the code and disagrees with the purchase sheet (above).
4. The website's BOM section carried **LED matrix eyes** ($120) and a **13-inch torso touch-screen**. Neither appears in the canonical MuJoCo model; both appear only in a placeholder scene file. They are not in this BOM.

Separately, the previously published cost figure of $9,860 with a nine-category split does not correspond to any purchase record and should be retired in favour of Table 2. The "under $10,000" claim survives, but it is a claim about scenario S2 ($9,624), not about a build that includes a Jetson Thor.

5 Known gaps and uncertainty

The purchase sheet is a record of what was ordered, not a complete parts explosion. Seven categories of real hardware are on the robot and costed nowhere. They are estimated here, deliberately kept out of the headline totals, and labelled as estimates.

Uncosted item	**Qty**	**Low ($)**	**High ($)**
ORCA hand structural kit (printed frame + tendons + bearings + fingertips)	2	300	600
3D-printed shells and covers (PETG/ABS filament)	1	45	60
Custom distribution PCBs (base + body)	2	40	240
PCB populated parts (connectors, regulators, CAN transceivers)	1	60	200
USB-CAN adapter (bench / direct-drive path)	1	25	45
Hardware E-stop switch + contactor	1	30	80
Microphone (head)	1	10	30
Estimated range		**510**	**1,255**

Table 5: Uncosted items, with estimated ranges. These are *not* included in Tables 1 or 2.

The largest is the ORCA hand structure. The sheet buys 34 Feetech servos but no frame, tendon, bearings, tensioners or fingertip pads — the hand is tendon-driven and printed, and those parts are real. The custom distribution PCBs are unverifiable from the repository because no gerbers have been released under `hardware/`; the row exists on the sheet with no price and stays open here.

This matters for the headline claim. Adding the gap range to scenario S2 gives $10,134–$10,879, which is *above* $10,000 at both ends. The honest statement is therefore narrower than the one the project has been making: **the parts on the purchase record total $9,624 in scenario S2, and a fully costed build — including the hand structure, the distribution PCBs and the E-stop — lands just over $10,000.** Only scenario S3 (no onboard compute, $9,375) stays under the threshold once the gaps are carried. A prospective builder should budget the S2 number plus the gap range plus shipping and duties, which for the Chinese-vendor block are not trivial.

6 Implementation and usage

6.1 Regenerating this document

```
cd BOM
python3 tools/build_bom.py      # CSV -> figures/*.pdf, generated/*.tex, JSON
make                            # generated/*.tex -> mabel_bom.pdf
```

`build_bom.py` reads the four CSVs under `data/`, converts currency by Eq. 1, computes subsystem subtotals and all four scenarios, writes four vector figures and five LaTeX fragments, emits `generated/bom_summary.json`, and prints a summary table. It hard-codes no totals: changing a price in the CSV changes the abstract of this document.

6.2 How the parts compose

The BOM's sections mirror the robot's control decomposition, which is why the scenario arithmetic is meaningful rather than cosmetic. The swerve base, arms, hands, torso and neck are each owned by a driver in the hardware-abstraction layer and reached over one of three bus types (REV SPARK CAN, DAMIAO MIT-mode CAN, and 1 Mbit TTL for the hands and the Dynamixel neck). Sensing is a separate tree on USB 3 and does not participate in control. Compute sits above all of it behind a single wire contract. Removing a whole section — as scenarios S3 and S4 do — therefore removes a coherent subsystem, not an arbitrary slice of a spreadsheet.

6.3 The website mirror

`generated/bom_summary.json` is the artefact the project website reads. The public BOM page renders the same line items, subsystem shares and scenario totals as this document, from the same build. Editing prices in one place and publishing them in two is the point of the pipeline.

7 Limitations

- **Marketplace prices are not catalog prices.** The Amazon and Taobao lines are as-purchased and will drift. $5,124 of the total (40%) sits on listings that are not stably addressable, which is the main reason this document is dated rather than evergreen.
- **No shipping, duty or tax.** For the CNY block in particular, international shipping and duty are a material fraction and are entirely absent here.
- **One robot, no spares.** The tendon-driven hands are the highest-wear subsystem on the platform, and a working data-collection programme will consume spare servos, tendon and fingertips. Nothing here budgets for that.
- **The neck question is open.** §4 records a genuine conflict between the purchase record and the driver's model-number table. It is worth ten minutes at the robot with the bus scanner to settle.
- **Estimates are labelled but still estimates.** Every number in Table 5 is a judgement, not a receipt, and the range is wide on purpose.
- **Peer prices are heterogeneous.** Figure 3 mixes manufacturer-stated prices, retail listings and third-party reports, and commercial platforms are frequently inquiry-only. It supports an order-of-magnitude argument, not a precise ranking.

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